

CIFAR-10 noise-ratio sweep and 48-client validation

IN-replace non-IID FedAvg seed 42 20 rounds 5 local epochs clipping norm 5.0

Recorded configuration

	3-client sweep	48-client validation
Active / canonical clients	3 / 4	48 / 49
Dataset / server test	4-class CIFAR-10 / 2,000 examples	4-class CIFAR-10 / 2,000 examples
Privacy modes	Vanilla, Global-DP, Metric-privacy	Vanilla, Global-DP, Metric-privacy
Noise ratios	0.002500, 0.003333, 0.006250	0.002500, 0.003333, 0.006250

Noise calibration: $\text{ratio} = \text{noise multiplier} / \text{active clients}$; $\text{sigma} = \text{ratio} \times 5$

Ratio	3-client multiplier	48-client multiplier	Applied sigma
0.002500	0.007500	0.120000	0.012500
0.003333	0.009999	0.159984	0.016665
0.006250	0.018750	0.300000	0.031250

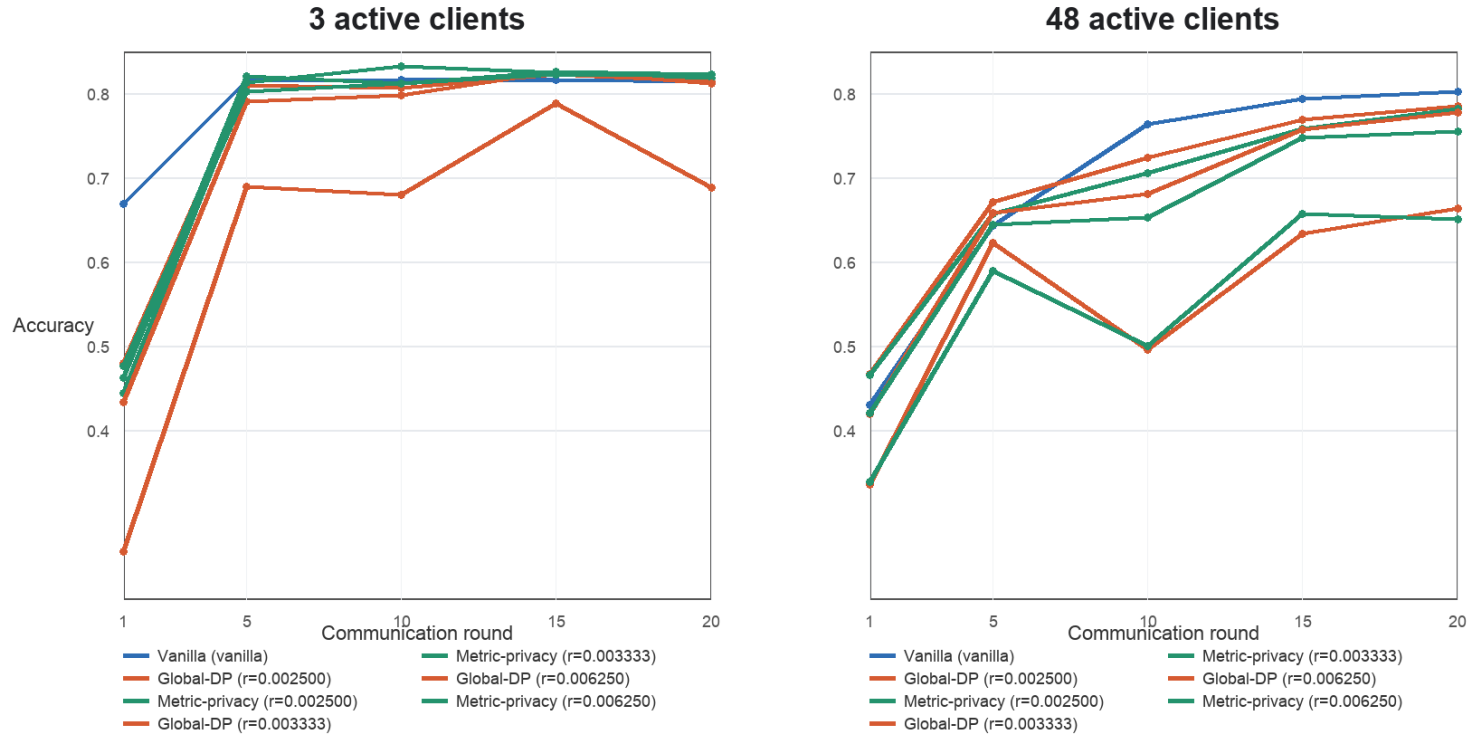
Final server-test metrics: 3 active clients

Privacy	Ratio	Multiplier	Accuracy	Macro F1	Log loss	Macro OVR AUC	Accuracy at rounds 1 / 5 / 10 / 15 / 20
Vanilla	–	0.010000	0.8135	0.8129	1.0808	0.9544	0.6690 / 0.8155 / 0.8160 / 0.8160 / 0.8135
Global-DP	0.002500	0.007500	0.8140	0.8146	0.8537	0.9535	0.4795 / 0.8100 / 0.8070 / 0.8230 / 0.8140
Metric-privacy	0.002500	0.007500	0.8185	0.8189	0.9047	0.9539	0.4445 / 0.8130 / 0.8320 / 0.8250 / 0.8185
Global-DP	0.003333	0.009999	0.8120	0.8132	0.8052	0.9519	0.4340 / 0.7900 / 0.7980 / 0.8260 / 0.8120
Metric-privacy	0.003333	0.009999	0.8225	0.8224	0.8547	0.9524	0.4630 / 0.8200 / 0.8115 / 0.8230 / 0.8225
Global-DP	0.006250	0.018750	0.6885	0.6943	1.2711	0.9095	0.2560 / 0.6890 / 0.6800 / 0.7880 / 0.6885
Metric-privacy	0.006250	0.018750	0.8225	0.8230	0.7941	0.9553	0.4765 / 0.8025 / 0.8115 / 0.8260 / 0.8225

Final server-test metrics: 48 active clients

Privacy	Ratio	Multiplier	Accuracy	Macro F1	Log loss	Macro OVR AUC	Accuracy at rounds 1 / 5 / 10 / 15 / 20
Vanilla	–	0.010000	0.8025	0.8016	0.5363	0.9503	0.4305 / 0.6435 / 0.7640 / 0.7940 / 0.8025
Global-DP	0.002500	0.120000	0.7855	0.7870	0.5967	0.9434	0.4670 / 0.6715 / 0.7235 / 0.7685 / 0.7855
Metric-privacy	0.002500	0.120000	0.7820	0.7826	0.6357	0.9391	0.4660 / 0.6575 / 0.7050 / 0.7580 / 0.7820
Global-DP	0.003333	0.159984	0.7775	0.7790	0.6516	0.9378	0.4195 / 0.6580 / 0.6810 / 0.7575 / 0.7775
Metric-privacy	0.003333	0.159984	0.7550	0.7565	0.7367	0.9288	0.4205 / 0.6445 / 0.6525 / 0.7470 / 0.7550
Global-DP	0.006250	0.300000	0.6635	0.6619	1.0295	0.8898	0.3360 / 0.6230 / 0.4955 / 0.6335 / 0.6635
Metric-privacy	0.006250	0.300000	0.6505	0.6457	1.0472	0.8778	0.3390 / 0.5900 / 0.5000 / 0.6570 / 0.6505

Server-test accuracy over communication rounds



Final-accuracy difference from vanilla (DP accuracy minus vanilla accuracy)

Ratio	3-client Global-DP	3-client Metric-privacy	48-client Global-DP	48-client Metric-privacy
0.002500	+0.0005	+0.0050	-0.0170	-0.0205
0.003333	-0.0015	+0.0090	-0.0250	-0.0475
0.006250	-0.1250	+0.0090	-0.1390	-0.1520

Source: `results/planned_runs/cifar/noise_sweep/` and `results/planned_runs/cifar/validation_48/`